Euclidean Geometry In Mathematical Olympiads

Problem 1.50: IMO 2013/4

geometry

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Nov 16, 2018, 3:08 PM • 2 #1

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This Problem came in IMO 2013 and is given as Problem 1.50 in the book Euclidean Geometry in Mathematical Olympiads by Evan Chen.

Hints

1. Add a Miquel point
2.Show that X,H,P are collinear , where P is said Miquel point
3. Do you see a pair of perpendicular lines?

Attachments:

Problem 1.50 (IMO 2013/4). Let ABC be an acute triangle with orthocenter H, and let W be a point on the side $\overline{BC}$, between B and C. The points M and N are the feet of the altitudes drawn from B and C, respectively. ω_1 is the circumcircle of triangle BWN and X is a point such that $\overline{WX}$ is a diameter of ω_1. Similarly, ω_2 is the circumcircle of triangle CWM and Y is a point such that $\overline{WY}$ is a diameter of ω_2. Show that the points X, Y, and H are collinear. **Hints:** 106 157 15 **Sol:** p.243

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Solution

Let K be the Miquel point , $i.e$, K lies on the circumcircle of ΔANM,ΔCWM,ΔBWN.
Let $\angle NWK = x$ and $\angle KWB = y$. Then, $\angle NWB = y - x$.

In quadrilateral $XDWB$,$\angle DXB$+$\angle DWB + \angle XDW + \angle XBW$ =360°
$\implies \angle XDW = 90^\circ + x$
Let $\angle BNW = \theta \implies \angle BNX = \angle WNH = 90^\circ - \theta$
In ΔNHD, $\angle DNH$+$\angle DHN + \angle NDH = 180^\circ$
$\implies 90^\circ - \theta + \angle DHN + 90^\circ + x = 180^\circ$
$\implies \angle DHN = \theta - x$
Let $\angle HNK = \phi \implies \angle KND = \angle HND - \angle HNK$
$= 90^\circ - \theta - \phi$
Now, $\angle HNK = \angle HMK = \phi$
In $\Delta MKH, \angle MKH + \angle KMH + \angle KHM = 180^\circ$
$\implies \angle MKH + \phi + \angle KHN + \angle NHM = 180^\circ$
$\implies \angle MKH + \phi + \theta - x + 180^\circ - 2\alpha$ (Here, $\angle BAC = 2\alpha$)
In $\Delta NKD, \angle NKD + \angle NDK + \angle KND = 180^\circ$
$\implies \angle NKD + 90^\circ + x + 90^\circ - \theta - \phi = 180^\circ$
$\implies \angle NKD = \theta + \phi - x$
$\implies$
$\angle NKD + \angle MKH + \angle NKM = \theta + \phi - x + 2\alpha + x - \theta - \phi + 180^\circ - 2\alpha$
$= 180^\circ$
$\implies X, K, H$ are collinear.
But , X, K, Y are also collinear ($\angle XKW$ + $\angle YKW = 90^\circ + 90^\circ = 180^\circ$)
$\implies X, Y, H$ are collinear.
Hence, Proved.

This post has been edited 1 time. Last edited by gautamlucky2002, Nov 16, 2018, 3:58 PM

Arthur.
133 posts

Nov 16, 2018, 5:03 PM • 2 #3

https://artofproblemsolving.com/community/c618937h1725224_problem_150_imo_20134
We already have a thread for IMO 2013/4 but it's perhaps my fault for not updating the problem discussion links thread

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Nov 16, 2018, 11:31 PM • 2 #4

No problem,Arthur. Even I should have first searched in the community before making this thread!

Kagebaka
3001 posts

May 25, 2019, 9:18 PM • 3 #6

Epic coaxiality finish

455264
74 posts

Oct 29, 2019, 2:42 PM • 1 #7

Well, just proving that $A, P,$ and W are collinear works

This post has been edited 2 times. Last edited by 455264, Oct 29, 2019, 2:43 PM